

Lawvere–Yanofsky Translation of Willard’s SJAS

R6 of the Refinement stage. Charter: ADR-0004-lawvere-sjas-translation.md.

Methodological order (binding). First prove G2 using Lawvere / Yanofsky techniques, with every hypothesis named. Only then determine where Willard’s constructions invalidate particular hypotheses of that limitative theorem.

Provenance. Claims about Willard’s corpus cite image-verified register rows in VERIFICATION.md or extraction records. Departures from Willard’s vocabulary are declared. This document does **not** re-prove Willard’s consistency theorems for $\text{IS}(\mathbf{A})$; it translates their logical structure into Lawvere / Yanofsky form and proves the translation theorems.

0. The complete result (what is proved)

Main Theorem (Self-justification in the Lawvere–Yanofsky scheme). Let α be a theory with the data of hypotheses H0 and H2 below (Lindenbaum objects, arithmetized provability). Then:

1. **(G2 in-scheme.)** If H0–H7 all hold, then $\alpha \vdash \text{Con}$ implies $\alpha \vdash \perp$. Under external consistency H8, therefore $\alpha \not\vdash \text{Con}$. (*Theorem 3.2.*)
2. **(Necessary conditions.)** If H0–H2, H6, H8 hold and $\alpha \vdash \text{Con}$, then at least one of H3–H5, H7 fails. (*Theorem 6.2 — obligatory breach.*)
3. **(Mechanism.)** For Willard’s Type-A system $\text{IS}(\mathbf{A})$, H1 holds (Group-3 is a Lawvere fixed point) while the **uniform** H4 arrow $\text{mp}_{\square}$ fails to lie in the affordable class $\text{Aff}(\text{IS}(\mathbf{A}))$ — Willard’s own uniformity failure, stated as a categorical missing arrow. (*Theorems 5.5–5.6.*)
4. **(Definition.)** α is *self-justifying* in this scheme iff H1, H8, and $\vdash \text{Con}$ hold and at least one of H3–H7 fails. (*Definition 6.1; necessity proved, sufficiency of the breach alone disproved by Beklemishev–Shamkanov.*)

That is the complete result. The sections prove it in order.

1. Lawvere / Yanofsky (the scheme)

We work in Yanofsky’s product form (Yanofsky 2022 Thms 6.4–6.5; cf. Lawvere 1969, Yanofsky 2003): terminal object and binary products suffice.

Definition 1.1 (Fixed point). $p : 1 \rightarrow Y$ is a fixed point of $\alpha : Y \rightarrow Y$ when $\alpha \circ p = p$.

Definition 1.2 (Representability). $g : A \rightarrow Y$ is *representable in* $f : A \times A \rightarrow Y$ when $\exists p : 1 \rightarrow A$ with $g = f \circ (p \times \text{id}_A) \circ i$ for $i : A \simeq 1 \times A$.

Theorem 1.3 (Diagonal form; Yanofsky 2022 Thm 6.4). If $\alpha : Y \rightarrow Y$ has no fixed point, then for every A and every $f : A \times A \rightarrow Y$ there is $g : A \rightarrow Y$ not representable in f .

Proof. Let $\Delta : A \rightarrow A \times A$ and $g := \alpha \circ f \circ \Delta$. If p represents g , then $f \circ \Delta \circ p$ satisfies $\alpha \circ (f \circ \Delta \circ p) = f \circ \Delta \circ p$, a fixed point. Contradiction. $\square$

Theorem 1.4 (Fixed-point form; Yanofsky 2022 Thm 6.5). If some $f : A \times A \rightarrow Y$ represents every $g : A \rightarrow Y$, then every $\alpha : Y \rightarrow Y$ has a fixed point.

Proof. For arbitrary α , the map $g := \alpha \circ f \circ \Delta$ is representable by some p ; then $f \circ \Delta \circ p$ fixes α . $\square$

Definition 1.5 (Lawvere situation). (A, Y, f) where f represents every $A \rightarrow Y$.

2. The Lawvere half: diagonal lemma

Setup 2.1. Let Lind^i be the Lindenbaum algebra of formulas with i free variables, modulo provable equivalence in the ambient theory. Define

$$f : \text{Lind}^1 \times \text{Lind}^1 \rightarrow \text{Lind}^0, \quad f(\mathcal{B}(x), \mathcal{H}(y)) = \mathcal{H}(\ulcorner \mathcal{B}(x) \urcorner).$$

Assume a diagonalisation function D on Gödel numbers with $D(\ulcorner \mathcal{B}(x) \urcorner) = \ulcorner \mathcal{B}(\ulcorner \mathcal{B}(x) \urcorner) \urcorner$, available as in the usual arithmetization (Yanofsky 2003; Mendelson).

Theorem 2.2 (Diagonal lemma as Lawvere fixed point). For any unary formula $\mathcal{E}(x)$, define $\Phi_{\mathcal{E}} : \text{Lind}^0 \rightarrow \text{Lind}^0$ by $\Phi_{\mathcal{E}}(\mathcal{P}) = \mathcal{E}(\ulcorner \mathcal{P} \urcorner)$. Then $\Phi_{\mathcal{E}}$ has a fixed point $\mathcal{C}$ with $\vdash \mathcal{C} \leftrightarrow \mathcal{E}(\ulcorner \mathcal{C} \urcorner)$.

Proof (fully expanded). Let $A = \text{Lind}^1$, $Y = \text{Lind}^0$, and f as above. Define $g : A \rightarrow Y$ by

$$g = \Phi_{\mathcal{E}} \circ f \circ \Delta,$$

so $g(\mathcal{B}(x)) = \mathcal{E}(\ulcorner \mathcal{B}(\ulcorner \mathcal{B}(x) \urcorner) \urcorner)$.

Let $\mathcal{G}(x) := \mathcal{E}(D(x))$. Then for any $\mathcal{B}(x)$,

$$g(\mathcal{B}(x)) = \mathcal{E}(\ulcorner \mathcal{B}(\ulcorner \mathcal{B}(x) \urcorner) \urcorner) = \mathcal{E}(D(\ulcorner \mathcal{B}(x) \urcorner)) = \mathcal{G}(\ulcorner \mathcal{B}(x) \urcorner) = f(\mathcal{B}(x), \mathcal{G}(y)).$$

Thus g is representable in f by the point $p : 1 \rightarrow A$ selecting $\mathcal{G}$. By Theorem 1.4, $\Phi_{\mathcal{E}}$ has a fixed point at

$$\mathcal{C} = \mathcal{G}(\ulcorner \mathcal{G}(x) \urcorner) = \mathcal{E}(D(\ulcorner \mathcal{G}(x) \urcorner)) = \mathcal{E}(\ulcorner \mathcal{C} \urcorner),$$

and the chain of equalities in Lind^0 is exactly $\vdash \mathcal{C} \leftrightarrow \mathcal{E}(\ulcorner \mathcal{C} \urcorner)$. $\square$

Corollary 2.3 (G1). Taking $\mathcal{E}(x) \equiv \neg \text{Der}(x)$ yields G with $\vdash G \leftrightarrow \neg \Box G$. Under external consistency / ω -consistency, G is unprovable. $\square$

Corollary 2.4 (Group-3 is Lawvere). Willard's Group-3 axiom is constructed by a Kleene / Rogers / Jeroslow fixed point over a substitution predicate (codified §5.1; Willard1993-TR Appendix A; Willard2005 Eqs. 7–8). That construction is Theorem 2.2 applied to a consistency-shaped $\mathcal{E}$. Therefore every affirmative Willard system **satisfies H1** below: it is a Lawvere situation for the endomorphism building Group-3. $\square$

3. G2 proved in the Lawvere / Yanofsky scheme

3.1 Hypothesis register

Id	Hypothesis	Content
H0	Ambient	Category with 1 and products; $Y = \text{Lind}^0$; $A = \text{Lind}^1$
H1	Lawvere representability	f of Setup 2.1 represents the g of Theorem 2.2 (diagonal lemma)
H2	Provability morphism	$\Box : Y \rightarrow Y$ interprets $\Phi \mapsto \text{Der}(\ulcorner \Phi \urcorner)$
H3	HB (1)	$\alpha \vdash \Phi \Rightarrow \alpha \vdash \Box \Phi$
H4	HB (2)	$\alpha \vdash (\Box \Phi \wedge \Box(\Phi \supset \Psi)) \supset \Box \Psi$ — uniformly in Φ, Ψ
H5	HB (3)	$\alpha \vdash \Box \Phi \supset \Box \Box \Phi$
H6	Π_1 -strength	α proves all Π_1^- theorems of PA (Willard2001 Thm A.1)
H7	Internal propositional packaging	α proves the propositional / first-order steps listed as lemmas in §3.2
H8	External consistency	$\alpha \not\vdash \perp$

Definition 3.1. A *Lawvere–HB $G2$ -situation* satisfies H0–H7.

H4 is stated with Willard’s uniformity: Willard2001 Thm A.1 says “Suppose **for any two sentences** Φ and Ψ ” (VERIFICATION.md, img at p. 44). The TR’s breach is exactly the failure of that uniformity (Willard1993-TR printed p. 12, img).

3.2 Packaging lemmas (making H7 finite and checkable)

Write $\text{Con} := \neg \Box \perp$. Assume (FP): $\alpha \vdash G \leftrightarrow \neg \Box G$, from Theorem 2.2 with $\mathcal{E}(x) = \neg \text{Der}(x)$.

Lemma 3.2.1. [H7, FP] $\alpha \vdash G \supset \neg \Box G$ and $\alpha \vdash \neg \Box G \supset G$. $\square$

Lemma 3.2.2. [H7, FP] $\alpha \vdash \Box G \supset \neg G$.

Proof. From (FP), $\alpha \vdash \Box G \supset \neg(\neg \Box G)$. But $\alpha \vdash \neg \Box G \leftrightarrow G$, so $\alpha \vdash \Box G \supset \neg G$. $\square$

Lemma 3.2.3. [H3, H4, H7] If $\alpha \vdash \Phi \supset \Psi$, then $\alpha \vdash \Box \Phi \supset \Box \Psi$.

Proof. H3 gives $\alpha \vdash \Box(\Phi \supset \Psi)$. H4 with $\Box \Phi$ and $\Box(\Phi \supset \Psi)$ yields $\Box \Psi$. $\square$

Lemma 3.2.4. [H3, H4, H5, H7, Lemmas 3.2.2–3.2.3] $\alpha \vdash \Box G \supset \Box \perp$.

Proof. From Lemma 3.2.2, $\alpha \vdash \Box G \supset \neg G$. By Lemma 3.2.3, $\alpha \vdash \Box \Box G \supset \Box \neg G$. By H5, $\alpha \vdash \Box G \supset \Box \Box G$. Compose: $\alpha \vdash \Box G \supset \Box \neg G$. From $\alpha \vdash \Box G$ and $\alpha \vdash \Box \neg G$, H4 applied to $\Box G$ and $\Box(\Box G \supset \perp)$ (since $\neg G$ is $G \supset \perp$) gives $\alpha \vdash \Box \perp$. Discharging the assumption $\Box G$ yields $\alpha \vdash \Box G \supset \Box \perp$. $\square$

Lemma 3.2.5. [H7, FP, Lemma 3.2.4] $\alpha \vdash \text{Con} \supset G$.

Proof. Lemma 3.2.4 contraposes to $\alpha \vdash \text{Con} \supset \neg \Box G$. With $\alpha \vdash \neg \Box G \supset G$, conclude $\alpha \vdash \text{Con} \supset G$. $\square$

3.3 The theorem

Theorem 3.2 (G2, Lawvere–HB form). Assume H0–H7. If $\alpha \vdash \text{Con}$, then $\alpha \vdash \perp$. Hence under H8, $\alpha \not\vdash \text{Con}$.

Proof.

(L) [H0, H1, H2] By Theorem 2.2, obtain (FP): $\alpha \vdash G \leftrightarrow \neg \Box G$. (*Sole use of Lawvere / Yanofsky.*)

(1)–(5) Lemmas 3.2.1–3.2.5 give $\alpha \vdash \text{Con} \supset G$ and $\alpha \vdash \Box G \supset \Box \perp$.

(6) [H3] Assume $\alpha \vdash \text{Con}$. Then $\alpha \vdash \Box \text{Con}$ and, from Lemma 3.2.5, $\alpha \vdash \Box(\text{Con} \supset G)$.

(7) [H4] From $\Box \text{Con}$ and $\Box(\text{Con} \supset G)$, H4 yields $\alpha \vdash \Box G$.

(8) From (7) and Lemma 3.2.4, $\alpha \vdash \Box \perp$, i.e. $\alpha \vdash \neg \text{Con}$. With $\alpha \vdash \text{Con}$, $\alpha \vdash \perp$.

(9) [H8] Therefore $\alpha \vdash \text{Con}$ contradicts H8. $\square$

Corollary 3.3 (Factorisation).

$$(H0+H1+H2) + (H3+H4+H5+H6+H7) \implies (\vdash \text{Con} \implies \vdash \perp).$$

Any single failure among H3–H7 blocks a tagged step. Failure of H1 blocks (L). $\square$

Corollary 3.4. Theorem 1.4 alone does not entail $\not\vdash \text{Con}$. G2 in this scheme is Theorem 3.2. $\square$

Remark 3.5 (Relation to Willard2001 Thm A.1). Theorem 3.2 is the Lawvere-factored form of Willard’s Theorem A.1 (status **stated-only** in the registry; justification by reduction to classical HB). H6 is Willard’s Π_1 -calibration; H3–H5 are his conditions (1)–(3) verbatim; H1 is the diagonal half he isolates as needing PA-strength for the fixed-point identity (Willard2001 footnote on the fixed-point / HB split, recorded in `refined-sjas.md` §2.1).

4. Alternate factorisations (declared, not primary)

Jeroslow + $\Box$ -contraction. Fixed point $G \leftrightarrow \Box(G \rightarrow \perp)$ plus $\text{copy}_\Box$ yields formalized G2 (Beklemishev–Shamkanov; ALSJAS AbstractG2). Different post-Lawvere package; Willard comparison uses §3’s register, not this one.

Kreisel model-chain. Yanofsky 2003 flags it; not developed here — Willard’s breach maps cleanly onto H4, not onto Kreisel’s truth-path diagonal.

5. Affordable arrows and Willard’s H4 failure

5.1 Ambient maps and affordability

Definition 5.1 (Ambient category $\overline{\mathcal{A}}$). Objects: Gödel numbers of formulas / proofs (as $\mathbb{N}$ where convenient), and the Lindenbaum objects A, Y of H0. Morphisms: partial recursive (arithmetically definable) maps on codes. Classical HB arguments are written externally in $\overline{\mathcal{A}}$.

Definition 5.2 (Affordable class $\text{Aff}(\alpha)$). $\text{Aff}(\alpha)$ is a **wide class of arrows** of $\overline{\mathcal{A}}$ (same objects; a designated subclass of morphisms). It is **not** assumed to be a subcategory: composition of

affordable arrows need not be affordable, because that composition is essentially $\text{mp}_\square$ (Definition 5.3), which Theorem 5.5 removes. Membership:

- An arrow m of $\overline{\mathcal{A}}$ lies in $\text{Aff}(\alpha)$ when α proves the **uniform** existence / totality statement defining m .
- If α proves only *instances* for fixed closed terms $\bar{c}$, the instance arrows $m_{\bar{c}} : 1 \rightarrow Z$ may lie in $\text{Aff}(\alpha)$ while the uniform m does not.
- Identity arrows are affordable ($\alpha \vdash \forall x x = x$).

This is the categorical reading of Willard’s fixed-constants vs uniform quantification (Willard1993-TR printed p. 12, img; same pattern at $\text{SUBST_i}(\bar{k}, y)$ for fixed $\bar{k}$).

Definition 5.3 (Uniform H4 arrow).

$$\text{mp}_\square : \square\Phi \times \square(\Phi \supset \Psi) \longrightarrow \square\Psi$$

in $\overline{\mathcal{A}}$ sends a pair of proof-codes (x, y) to a proof-code z of Ψ when such a z exists. **H4 holds relative to $\text{Aff}(\alpha)$** iff $\text{mp}_\square \in \text{Aff}(\alpha)$, i.e. iff α proves

$$\forall\Phi\forall\Psi (\square\Phi \wedge \square(\Phi \supset \Psi)) \supset \square\Psi.$$

Definition 5.4 (Degenerate / instance H4 arrows). For fixed closed proof-codes $\bar{x}, \bar{y}$, the instance arrow $\text{mp}_{\bar{x}, \bar{y}} : 1 \rightarrow \square\Psi$ may lie in $\text{Aff}(\alpha)$ even when $\text{mp}_\square$ does not.

5.2 The missing-arrow theorem for IS(A)

Theorem 5.5 (H4 fails as a missing Aff-arrow). Let $\alpha = \text{IS}(A)$ for nice A , in Willard’s Type-A / semantic-tableaux configuration. Then:

1. $\text{mp}_\square \notin \text{Aff}(\alpha)$ (uniform H4 fails);
2. for fixed constants $\bar{x}, \bar{y}$, instance arrows of Definition 5.4 may lie in $\text{Aff}(\alpha)$ (degenerate H4 holds);
3. H1 holds for α (Group-3 / Lawvere fixed point is an axiom).

Proof. (1)–(2) are Willard’s statement, image-verified at Willard1993-TR printed p. 12 (VERIFICATION.md; extraction willard1993-tr.md §3.7c):

“For the case of $\text{IS}(A)$, the difficulty is that **only in the *degenerate case* where $\mathbf{x}$ and $\mathbf{y}$ are *fixed constants*** can $\text{IS}(A)$ prove the Hilbert-Bernays condition that if $\mathbf{x}$ is a proof of α and $\mathbf{y}$ is a proof of $\alpha \supset \beta$ then there exists some z that proves β .”

Under Definitions 5.2–5.4 that sentence is exactly (1)–(2). No inference beyond the translation into those definitions is used.

(3) is Corollary 2.4. $\square$

Theorem 5.6 (Evasion mechanism). Relative to $\text{Aff}(\text{IS}(A))$, Theorem 3.2 fails at H4: Lemmas 3.2.3–3.2.4 and step (7) invoke $\text{mp}_\square \notin \text{Aff}(\alpha)$. The Lawvere half (L) remains. Therefore $\vdash \text{Con} \Rightarrow \vdash \perp$ is not forced by Theorem 3.2 inside $\text{Aff}(\text{IS}(A))$.

Proof. Theorem 5.5(1) + Corollary 3.3. $\square$

Theorem 5.7 (Obligatory breach, Aff-form). Suppose α satisfies H0–H2, H6, H8 and $\alpha \vdash \text{Con}$, with the affirmative corpus theorems applying. Then α is not a Lawvere–HB G2-situation relative to $\text{Aff}(\alpha)$: at least one of H3–H5, H7 fails there. For Type-A $\text{IS}(A)$, the failed hypothesis is H4.

Proof. If H0–H7 all held relative to $\text{Aff}(\alpha)$, Theorem 3.2 would give $\alpha \vdash \perp$, contradicting H8. For $\text{IS}(\mathbf{A})$, Theorem 5.5 names H4. $\square$

5.3 Filled hypothesis register for Type-A $\text{IS}(\mathbf{A})$

Id	Step(s)	Status relative to $\text{Aff}(\text{IS}(\mathbf{A}))$	Provenance
H0	ambient	holds	Setup
H1	(L)	holds	Cor. 2.4; codified §5.1
H2	$\square$	holds	arithmetization in corpus
H3	Lem. 3.2.3, Thm 3.2(6)	not the named breach	TR names condition (2) specifically
H4	Lem. 3.2.3–3.2.4, Thm 3.2(7)	fails (uniform)	Thm 5.5; TR p. 12
H5	Lem. 3.2.4	not the named breach	—
H6	Π_1	holds by design	Thm A.1 calibration; TR Remark 7
H7	packaging	fails insofar as it invokes uniform H4	Def. 5.2
H8	consistency	holds	TR Prop. 1 / 2001 Thm 3.4 (full)

5.4 Margin as quantitative affordability

Definition 5.8 (Margin; Willard2011 Def. 4.5,). $M(\xi) = \inf_{\beta} (\text{Log}(q_{\beta}) - \sharp(\beta))$.

Theorem 5.9 (Margin witnesses failure of the G2 composite). If $M(\xi) \geq 2$, then by Willard2011 Lemma 4.6 (status **sketch**, proof “kept brief and informal”; at pp. 14–15) the configuration with its self-referential axiom is a consistent self-justifying system. In the present vocabulary: $M(\xi) \geq 2$ implies that the composite arrow of Theorem 3.2 producing $\perp$ from Con is not in $\text{Aff}(\xi)$, while H1 and $\vdash \text{Con}$ can hold.

Proof content. Eqs. (19)–(20) place the fixed point at margin exactly 1; demanding margin ≥ 2 excludes every inconsistent β from matching that cost (**R3-the-margin.md** §6, page-image verified). That exclusion is the absence of the contradiction-producing arrow from $\text{Aff}(\xi)$. $\square$

Status note. Lemma 4.6 is **sketch** in the registry; Theorem 5.9 inherits that status for the quantitative half. Theorem 5.5 (uniform H4 failure for $\text{IS}(\mathbf{A})$) does **not** inherit it — it rests on Willard’s explicit statement, not on the margin lemma.

6. Necessary conditions for self-justification

Definition 6.1 (Self-justifying situation). A theory α with H0, H2 is *self-justifying* when:

	Condition
(SJ1)	H1 — Lawvere / Group-3 available
(SJ2)	H8 — consistent
(SJ3)	$\alpha \vdash \text{Con}$ — self-assertion of consistency
(SJ4)	at least one of H3–H7 fails in $\text{Aff}(\alpha)$

Theorem 6.2 (Necessity of (SJ4)). Assume H0–H2, H6, and (SJ1)–(SJ3). Then (SJ4) holds.

Proof. Suppose not: H3–H7 all hold in $\text{Aff}(\alpha)$. Then H0–H7 hold, so Theorem 3.2 yields $\alpha \vdash \perp$, contradicting (SJ2). $\square$

Theorem 6.3 (Insufficiency of (SJ4) alone). (SJ4) does not imply (SJ3). Beklemishev–Shamkanov’s system S fails formalized G2 (a post-Lawvere structural hypothesis in the §4.1 packaging) but does not prove $\neg \Box \perp$ (their §6, img in VERIFICATION.md). $\square$

Theorem 6.4 (Type-A instance). For $\alpha = \text{IS}(A)$ (nice A), (SJ1) holds by Corollary 2.4; (SJ2) and (SJ3) hold by Willard’s Proposition 1 / Theorem 3.4 (**full**); (SJ4) holds by Theorem 5.5 (H4 fails). Therefore $\text{IS}(A)$ is self-justifying in the sense of Definition 6.1, and the evasion mechanism is Theorem 5.6. $\square$

Theorem 6.5 (Conventional configurations are not self-justifying). If ξ is a conventional configuration satisfying H0–H7 in $\text{Aff}(\xi)$ (Willard2011 p. 21 fn. 11, img: “will satisfy the Hilbert-Bernays derivability conditions... automatically inconsistent”), then (SJ3) is incompatible with (SJ2) by Theorem 3.2. $\square$

7. Completion audit

Requirement	Evidence	Status
Prove G2 using Lawvere/Yanofsky techniques	§3: Thm 2.2 (expanded) + Lemmas 3.2.1–3.2.5 + Thm 3.2	proved
Named hypothesis register	§3.1 H0–H8; filled for Type-A in §5.3	done
Translate Willard into that framework	§§5–6: Aff, missing Aff-arrow, SJ conditions	proved
Necessary conditions for self-justification	Def 6.1; Thm 6.2 (necessity of breach)	proved
Mechanism of G2 evasion (category-theoretic)	Defs 5.1–5.4; Thms 5.5–5.7	proved
Type-A instance	Thm 6.4, citing corpus full consistency	proved (translation; consistency cited not re-derived)
Image-verified Willard H4 quote	VERIFICATION.md row TR p. 12	verified
Image-verified Thm A.1 uniformity	VERIFICATION.md row 2001 p. 44	verified

Requirement	Evidence	Status
Re-prove Willard’s consistency of $\text{IS}(\mathbf{A})$ from scratch	out of scope (codification owns it)	cited, not re-proved
Kreisel secondary register	§4	deferred (not required for the Main Theorem)
Mechanised Lean of Thm 3.2	optional	not required for the Main Theorem

AAR. The complete result stated in §0 is proved. Lawvere supplies only the fixed point; Willard’s Type-A evasion is the absence of the uniform $H4$ arrow $\text{mp}_{\square}$ from $\text{Aff}(\text{IS}(A))$, while $H1$ remains. Self-justification is exactly (SJ1)–(SJ4). The quantitative margin (Thm 5.9) is an additional sufficient witness at registry status **sketch** and is not required for the Main Theorem’s Type-A case, which rests on Willard’s explicit uniformity statement.

References

- Lawvere, F. W. (1969/2006). TAC Reprints 15.
- Yanofsky, N. S. (2003). BSL 9(3); arXiv:math/0305282.
- Yanofsky, N. S. (2022). *Working Category Theorist*, Ch. 6.
- Willard1993-TR printed p. 12 ($H4$ uniformity failure); Willard2001 Thm A.1 p. 44; Willard2011 Def. 4.5 / Lemma 4.6 — as in VERIFICATION.md.
- `refined-sjas.md`, `R3-the-margin.md`, ADR-0002, ADR-0004.